

Section: Particles and Waves

Topic: Radiation, The Inverse Square Law

A radioactive source is used in an experiment to investigate the inverse square law.

(a)

State what is meant by the inverse square law

✓ The inverse square law states that the **intensity** of radiation is **inversely proportional to the square of the distance** from the source:

$$I \propto \frac{1}{d^2}$$

This means:

- If the distance from the source **doubles**, the intensity **falls to one quarter**.

(b)

Calculate the distance from the source when the intensity is 1.2 units

Given:

- At $d_1 = 0.30 \text{ m}$, intensity $I_1 = 4.8$
- At $d_2 = ?$, intensity $I_2 = 1.2$

Use the inverse square law ratio:

$$\frac{I_1}{I_2} = \left(\frac{d_2}{d_1} \right)^2 \Rightarrow \frac{4.8}{1.2} = \left(\frac{d_2}{0.30} \right)^2$$

$$4 = \left(\frac{d_2}{0.30} \right)^2 \Rightarrow \frac{d_2}{0.30} = \sqrt{4} = 2 \Rightarrow d_2 = 0.30 \times 2 = 0.60 \text{ m}$$

 Revision Tips

- The **inverse square law** applies to radiation, light, sound, gravity, and electric fields
- Use ratio form:

$$\frac{I_1}{I_2} = \left(\frac{d_2}{d_1} \right)^2$$

to avoid needing constants

- Always **square root** carefully when solving for distance
- If intensity falls by a factor of 4, distance has **doubled**